

4.1

$$(a) R^2 = \frac{SSR}{SST} = \frac{375,47}{\sum (y_i - \bar{y})^2} = \frac{375,47}{\sum y_i^2 - N\bar{y}^2} = \frac{375,47}{7825,94 - 20 \times 19,21^2} = 0,8429$$

$$(b) \hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{SSE}{18} = \frac{(1-R^2) \times SST}{18} = \frac{0,2089 \times 7825,94}{18} = 8,4249$$

$$(c) R^2 = \frac{SSR}{SST} = \frac{\sum (y_i - \bar{y})^2 - \sum \hat{e}_i^2}{\sum (y_i - \bar{y})^2} = \frac{631,63 - 182,85}{631,63} = 0,7105$$

4.3

$$(a) \hat{y}_0 = 1,2 + 0,8 \times 4 = 4,4$$

$$(b) x_0 = 4, \bar{x} = \frac{5}{5} = 1, \hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3,6}{3} = 1,2$$

$$\Rightarrow se(f) = \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]} = \sqrt{1,2 \times \left[1 + \frac{1}{5} + \frac{9}{10} \right]} = 1,58745$$

(c)

a 95% prediction interval for y_0 : $[\hat{y}_0 - t_{(3, 0,025)} se(f), \hat{y}_0 + t_{(3, 0,025)} se(f)]$
 $\Rightarrow [4,4 - 3,1824 \times 1,58745, 4,4 + 3,1824 \times 1,58745]$
 $= [-0,6520, 9,4520]$

(d)

a 99% prediction interval for y_0 : $[\hat{y}_0 - t_{(3, 0,005)} se(f), \hat{y}_0 + t_{(3, 0,005)} se(f)]$
 $\Rightarrow [4,4 - 5,8409 \times 1,58745, 4,4 + 5,8409 \times 1,58745]$
 $= [-4,8722, 13,6722]$

(e)

$$x_0 = \bar{x} = 1, \hat{y}_0 = 1.2 + 0.8x = 2, \hat{\sigma}^2 = 1.2$$

$$\Rightarrow se(f) = \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]} = \sqrt{1.2 \times \left[1 + \frac{1}{5} + \frac{0}{10} \right]} = 1.2$$

a 95% prediction interval for y_0 : $[\hat{y}_0 - t_{(3, 0.025)} se(f), \hat{y}_0 + t_{(3, 0.025)} se(f)]$

$$\Rightarrow [2 - 3.1824 \times 1.2, 2 + 3.1824 \times 1.2]$$
$$= [-1.8189, 5.8189]$$

The width of part (c) = $9.4520 - (-0.6520) = 10.104$

The width of part (e) = $5.8189 - (-1.8189) = 7.6378$

The prediction interval is narrower for $x = \bar{x}$ compared to $x = 4$. This is because when we set x to a value not in the sample range ($x_i = 3, 2, 1, -1, 0$), such as $x = 4$, the regression model's uncertainty in predicting at this value is higher, which leads to a wider prediction interval.

4.25

(a)

$$\widehat{\ln(\text{PRICE})} = 4.393866 + 0.036044 \text{ SQFT}$$

(0.043288) (0.001486)

If the value of SQFT is equal to zero, the model predicts its expected value of $\ln(\text{PRICE})$ would be 4.393866.
(the price would be $e^{4.393866} \times 1000 = 80950$)

When SQFT increases by one, the price of the house in Baton Rouge is estimated to increase by $(100 \times 0.036044) = 3.6044$ percent on average.

$$\overline{\text{SQFT}} = \bar{X} = 27,28212, \quad \overline{\text{PRICE}} = 250,2369$$

$$\widehat{\ln(\text{PRICE})} = 4.393866 + 0.036044 \times 27,28 = 5.377$$

$$\widehat{\text{PRICE}} = e^{5.377} = 216,40$$

At the sample mean, the slope is $\frac{dy}{dx} = B_2 \cdot \hat{y} = 0.036044 \times 216,4 = 7,8$
and the elasticity is $B_2 \cdot \bar{X} = 0.036044 \times 27,28 = 0,98$

(b)

$$\widehat{\ln(\text{PRICE})} = 2.04965 + 1.02483 \ln(\text{SQFT})$$

(0.157997) (0.04839)

When $\ln(\text{SQFT})$ is zero, the expected value of $\ln(\text{PRICE})$ is 2.04965.

When SQFT increases by 1%, the expected price will increase by 1.02483%.

At the sample mean, the slope is $B_2 \frac{\bar{Y}}{\bar{X}} = 1.02483 \times \frac{250,2369}{27,28212} = 9.3999$
the elasticity is $B_2 = 1.02483$.

$$(C) \widehat{PRICE} = \underset{(13,0882)}{-115,4236} + \underset{(0,4442)}{13,4029} SQFT$$

Calculate "generalized" R^2 :

log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$

$$\hat{y} = e^{b_1 + b_2 x}$$

$$R_g^2 = [\text{corr}(y, \hat{y})]^2 = r_{y\hat{y}}^2$$

log-log model

$$\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$$

$$\hat{y} = e^{\alpha_1 + \alpha_2 \ln(x)}$$

$$R_g^2 = r_{y\hat{y}}^2$$

linear model

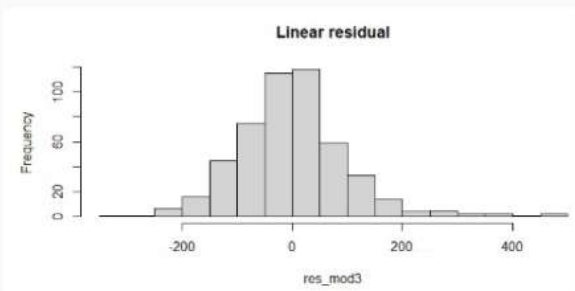
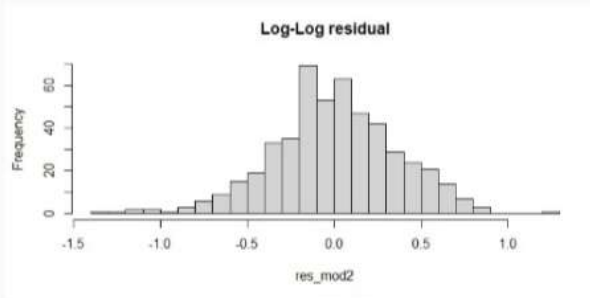
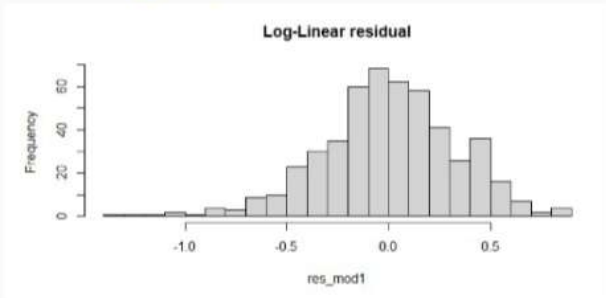
$$PRICE = \delta_1 + \delta_2 SQFT + e$$

$$\hat{y} = \delta_1 + \delta_2 x$$

$$R_g^2 = r_{y\hat{y}}^2$$

Model	R^2	R_g^2
log-linear	0.5417259	0.6621612
log-log	0.4738445	0.6445084
linear	0.6413169	0.6413169

(d)



```
> jarque.test(res_mod1)

Jarque-Bera Normality Test

data: res_mod1
JB = 26.679, p-value = 1.61e-06
alternative hypothesis: greater

> jarque.test(res_mod2)

Jarque-Bera Normality Test

data: res_mod2
JB = 14.279, p-value = 0.0007933
alternative hypothesis: greater

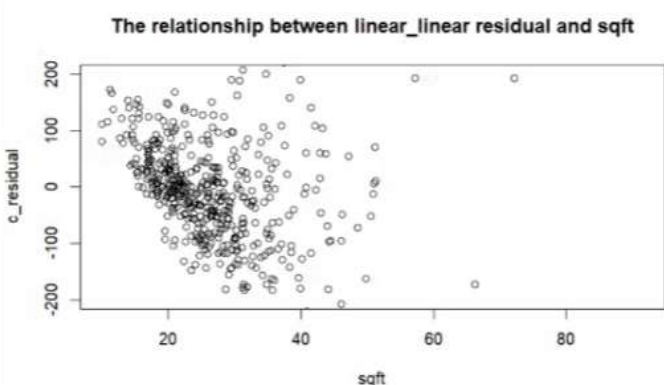
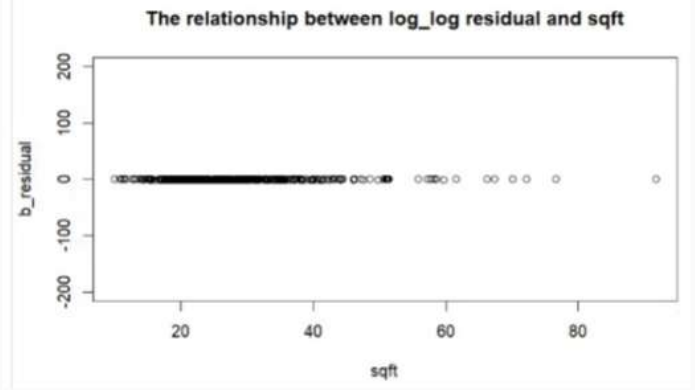
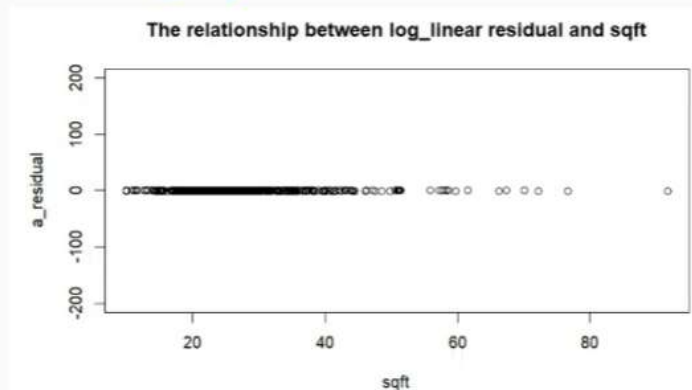
> jarque.test(res_mod3)

Jarque-Bera Normality Test

data: res_mod3
JB = 221.11, p-value < 2.2e-16
alternative hypothesis: greater
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The JB statistics' P-value is smaller than 0.05 in all the models, so we can conclude that these residuals do not follow the assumption of normality.

(e)



The residual fluctuations of log-linear and log-log are smaller than linear-linear. Given SQFT, log-linear and log-log satisfy that the expected value of residual is zero. The linear-linear model doesn't satisfy this condition.

(f)

$$\begin{aligned}\text{log-linear: } \hat{y} &= e^{B_1 + B_2 X} \Rightarrow (\hat{y}_0) \times 1000 = (e^{4.393866 + 0.036044 \times 27}) \times 1000 = 214233.6 \\ &\quad \text{(dollars)} \\ \text{log-log: } \hat{y} &= e^{\alpha_1 + \alpha_2 \ln(X)} \Rightarrow (\hat{y}_0) \times 1000 = (e^{2.04965 + 1.02483 \times \ln(27)}) \times 1000 = 227538.6 \\ &\quad \text{(dollars)} \\ \text{linear-linear: } \hat{y} &= \delta_1 + \delta_2 X \Rightarrow (\hat{y}_0) \times 1000 = (-115.4236 + 13.4029 \times 27) \times 1000 = 246455.7 \\ &\quad \text{(dollars)}\end{aligned}$$

(g)

$$\begin{aligned}\text{A 95\% prediction interval for log-linear model: } &[e^{\hat{y} \pm t_{0.025}(498) \cdot \text{se}(f)}] \times 1000 \\ &= [109768, 418116] \text{ (dollars)}\end{aligned}$$

$$\begin{aligned}\text{A 95\% prediction interval for log-log model: } &[e^{\hat{y} \pm t_{0.025}(498) \cdot \text{se}(f)}] \times 1000 \\ &= [111141, 465841] \text{ (dollars)}\end{aligned}$$

$$\begin{aligned}\text{A 95\% prediction interval for linear model: } &[\hat{y} \pm t_{0.025}(498) \cdot \text{se}(f)] \times 1000 \\ &= [44277, 448634] \text{ (dollars)}\end{aligned}$$

(h)

I would use the log-linear model because it has the highest generalized R^2 .